

[illegible]

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/*****
1) CREATE FORMATS
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/*formats*/
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proc format;
  VALUE JHU
    0      = 'Not Frail/Robust Health'
    1-2    = 'Pre-Frail'
    3-5    = 'Frail'
    .      = 'Missing' ;
  VALUE FESI
/*    0      = 'Missing'*/
    16-19  = 'Low concern'
    20-27  = 'Moderate concern'
    28-64  = 'High concern' ;
  VALUE MOCA
    26-30   = 'Normal'
    18-25   = 'Mild cognitive impairment'
    10-17   = 'Moderate cognitive impairment'
    0-9     = 'Severe cognitive impairment'
    .       = 'Missing';
  VALUE TMOCA
    19-22   = 'Normal'
    13-18   = 'Mild cognitive impairment'
    7-12    = 'Moderate cognitive impairment'
    1-6     = 'Severe cognitive impairment'
    0       = 'Missing';
  VALUE YesNofmt
    0       = 'No'
    1       = 'Yes' ;
  VALUE Callahan
    0-3     = 'Impairment Likely'
    4-6     = 'Not Impaired'
    .       = 'Missing';
  VALUE ADL
    0-23    = 'Not fully independent'
    24      = 'Independent' ;
  VALUE ASSESS
    0       = 'Not at risk'
    1-3     = 'At risk of Falls';
  VALUE FALLS
    0       = 'No Falls'
    1       = '1'
    2       = '2'
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3      = '3'
4      = '4'
5      = '5 or more';
/* VALUE VISIT*/
/* 1      = 'Baseline'*/
/* 2      = 'Follow-up 1'*/
/* 3      = 'Follow-up 2'*/
run;

/*****
2) IMPORT THE FILES AND CREATE ONE DATASET
*****/

%global dataset_list; /* Create a global macro variable to store dataset names */

%macro import_all_excels(dir=);
/* Use PIPE to get a list of all Excel files in the directory */
filename filelist pipe %sysfunc(quote(dir /b "&dir.\*.xlsx"));

/* Read file names into a dataset */
data file_list;
length filename $256 fullpath $512;
infile filelist truncover;
input filename $256.;
fullpath = cats("&dir.\", filename); /* Construct full file path */
run;

/* Reset dataset list */
%let dataset_list=;

/* Loop through each file and import it */
data _null_;
set file_list;
call execute(cats('%import_single_file(file=', fullpath, ')'));
run;

/* Clear the filename reference */
filename filelist clear;
%mend import_all_excels;

%macro import_single_file(file=);
/* Extract filename without extension for dataset name */
%let dataset_name = %sysfunc(scan(%sysfunc(scan(&file., -1, \)), 1, .));

/* Append dataset name to the global dataset list */
%if %symexist(dataset_list) %then %let dataset_list = &dataset_list work.&dataset_name;
%else %let dataset_list = work.&dataset_name;

/* Import the Excel file into WORK with filename as dataset name */
proc import out=work.&dataset_name
datafile="&file."
dbms=xlsx replace;
getnames=yes;
run;

/* Ensure numeric values and rename */
data work.&dataset_name;
set work.&dataset_name;

/* Convert Hospital_stays and ER to numeric if needed */
if vtype(Hospital_stays) = 'C' then Hospital_stays_num = input(Hospital_stays, best32.);
else Hospital_stays_num = Hospital_stays;

if vtype(ER) = 'C' then ER_num = input(ER, best32.);
else ER_num = ER;

/* Drop original variables and rename new ones */
drop Hospital_stays ER;
rename Hospital_stays_num = Hospital_stays
ER_num = ER;
run;
%mend import_single_file;

%macro merge_all_datasets;
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%if &dataset_list ne %then %do;
  data merged_data; /*THE FINAL DATASET*/
    merge &dataset_list;
    by patient_id visit_number;
  run;
%end;
%else %do;
  %put WARNING: No datasets were imported. Merge not executed.;
%end;

%mend merge_all_datasets;

/* execute the macros */
%import_all_excels(dir=DA\Ahmad\Aging and Fragility\);
%merge_all_datasets;

/*****
3) CREATE ANALYTIC COHORT FOR THIS REPORT
*****/

/*EDIT SOME MISSING/INCORRECT DATA AND CREATE A NEW AGE AND AGE-GROUP AT 9/11 VARIABLE*/

data REDCAP_&todayDate;
  retain Patient_ID visit_number Date_of_Birth Age_911 age_Group_at911
    gender Weight_lb Height Height_inches
    Falls_Assessment Frailty_Score_not_JHU JHU_score
    Hospital_stays
    ER ADL FESI MOCA TMOCA Callahan;

  set merged_data;

  /* Handle invalid ADL and MOCA values */
  if ADL in (-80, -74, -73) then ADL = .;
  if MOCA = 31 then MOCA = 30;

  /* Ensure Date_of_Birth is a valid SAS date */
  if not missing(Date_of_Birth) then Age_911 = INT(YRDIF(Date_of_Birth, '11SEP2001'd, 'ACTUAL'));
  else Age_911 = .; /* Explicitly set missing if Date_of_Birth is missing */

  /* Assign age group at 9/11*/
  select;
    when (missing(Age_911)) age_Group_at911 = 'Unknown'; /* Handle missing values properly */
    when (Age_911 < 18) age_Group_at911 = '< 18';
    when (Age_911 < 25) age_Group_at911 = '18-24';
    when (Age_911 < 35) age_Group_at911 = '25-34';
    when (Age_911 < 50) age_Group_at911 = '35-49';
    when (Age_911 < 55) age_Group_at911 = '50-54';
    when (Age_911 < 65) age_Group_at911 = '55-64';
    when (Age_911 < 75) age_Group_at911 = '65-74';
    otherwise age_Group_at911 = '75+';
  end;

run;

/*APPLY FORMAT TO USE THE BELOW DATASET FOR RUNNING REPORTS*/

data REDCAP_data;
  retain patient_id visit_number;
  format
    JHU_Score JHU.
    Frailty_Score_not_JHU JHU.
    FESI FESI.
    MOCA MOCA.
    TMOCA TMOCA.
    Falls_Assessment ASSESS.
    Callahan Callahan.
    ADL ADL.
  /* visit_number visit. */
  ;
  set cohort_&todayDate;
run;

/*****
4) GENERATE FREQUENCY RTF FILES
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/*USE DATA TO INPUT TO AN EXCEL FILE*/
/*NOTE: DATA IS STRATIFIED BY THE DIFFERENT VISITS (BASELINE= VISIT1, FOLLOW-UP 1= VISIT 2, FOLLOW-UP 2= VISIT 3)*/

/*1) Overall data, not broken down by any of the Frailty measures (JHU or non-JHU)*/

ods rtf file="DA\Ahmad\Aging and Fragility\freqs_overall.rtf";

%let vars = age_Group_at911 gender JHU_Score Frailty_Score_not_JHU ADL Callahan MOCA TMOCA FESI Falls_Assessment ER Hospital_

proc freq data=REDCAP_data;
  title "Descriptives, freq";
  by visit_number;
  tables &vars
  / list missing chisq expected;
run;

ods rtf close;

/*2) DATA BROKENDOWN BY JHU responsesnse*/

ods rtf file="DA\Ahmad\Aging and Fragility\freqs_JHU.rtf";

proc freq data=REDCAP_data (where = (JHU_Score ne .));
  by visit_number;
  tables gender*JHU_Score
         Frailty_Score_not_JHU*JHU_Score
         ADL*JHU_Score Callahan*JHU_Score MOCA*JHU_Score TMOCA*JHU_Score
         FESI*JHU_Score Falls_Assessment*JHU_Score
         ER*JHU_Score Hospital_stays*JHU_Score
  / list missing chisq expected;
run;
ods rtf close;

/*3) DATA BROKENDOWN BY Non-JHU responses*/

ods rtf file="DA\Ahmad\Aging and Fragility\freqs_nonJHU.rtf";

proc freq data=REDCAP_data (where = (Frailty_Score_not_JHU ne .));
  by visit_number;
  tables gender*Frailty_Score_not_JHU
         JHU_Score*Frailty_Score_not_JHU
         ADL*Frailty_Score_not_JHU Callahan*Frailty_Score_not_JHU MOCA*Frailty_Score_not_JHU TMOCA*Frailty_Score_not_JHU
         FESI*Frailty_Score_not_JHU Falls_Assessment*Frailty_Score_not_JHU
         ER*Frailty_Score_not_JHU Hospital_stays*Frailty_Score_not_JHU
  / list missing chisq expected;
run;

ods rtf close;

/*****
END END END END END END END END END END END END END END END
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/*USE DATA THE ABOVE 3 RTF FILES AND INPUT INTO AN EXCEL FILES TO SEND TO DR. KO ET AL. */

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